A064 v0.2.0 — PKLSA-Initiated Euler Flow Separatrix Candidate and Material-Transport Falsifier

SST Workbench numerical research prototype

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Abstract

An evolving, periodic, incompressible 3-D Euler numerical lane replaces the repeated frozen velocity fields in A064 v0.1.1. Original, unqualified 3-D KnotPlot/PKLSA centerline coordinates of 3_1, 5_2 and 6_1 are used only as initial-condition shapes. The workflow independently discovers enstrophy-level candidate boundaries before auditing normal relative flux and tracer crossing. The findings concern the selected model-dependent level sets, not global invariant manifolds and not physical SST evidence. Framework v1.0.7 FREEZE/CERTIFY has not run.

1 Source and evidence classification

Input centerlines: three raw native knot polygons, retained with SHA-256 hashes. E010 v0.3.1/E011 v0.3.0 certify static geometry at their own admitted provider anchors; these bundled native polylines have *not* been mapped to E011 STATIC_READY anchor IDs. E011 additionally reports 3_1 and 5_2 cross-provider robust and 6_1 cross-provider sensitive in static metrics. No physical dynamical velocity field has been imported.

All values in the numerical solver use *dimensionless computational units*. The NPZ uses an arbitrary $L_0 = 1\,\mathrm{m}$, $T_0 = 1\,\mathrm{s}$ bookkeeping convention for compatibility; this is not a measurement nor calibration of r_c or $\mathbf{v}_{\circlearrowleft}$.

2 Equations and numerical scheme

For an inviscid incompressible Euler fluid without external forces:

$$\nabla \cdot \mathbf{u} = 0, \qquad \partial_t \mathbf{u} + (\mathbf{u}\cdot\nabla)\mathbf{u} = -\frac{1}{\rho}\nabla p, \tag{1}$$

$$\boldsymbol{\omega} = \nabla \times \mathbf{u}, \qquad \partial_t \mathbf{u} = \mathsf{P}(\mathbf{u}\times\boldsymbol{\omega}), \tag{2}$$

where P is the Leray projection onto solenoidal fields. Spatial differentiation uses periodic 3-D Fourier modes with strict 2/3 truncation; time stepping uses RK4 and a CFL limit of 0.16. The initial knotted tube is a Gaussian-smoothed line-vorticity distribution projected into a divergence-free velocity field. It is a *model assumption*, not a unique physical PKLSA reconstruction.

The solver monitors energy $E = \frac{1}{2}\langle|\mathbf{u}|^2\rangle$, helicity $H = \langle \mathbf{u}\cdot\boldsymbol{\omega}\rangle$, divergence and

$$\frac{D|\boldsymbol{\omega}|^2}{Dt} = 2\omega_i\omega_j\partial_j u_i\,, \quad \frac{D}{Dt} = \partial_t + \mathbf{u}\cdot\nabla. \tag{3}$$

For a discovered level set $F(\mathbf{x},t) = |\boldsymbol{\omega}|^2 - \lambda$, its relative normal speed is

$$V_n^{\mathrm{rel}} = \frac{\partial_t F + \mathbf{u}\cdot\nabla F}{|\nabla F|}. \tag{4}$$

A vanishing value is necessary for this level set to be material; $\nabla\cdot\mathbf{u} = 0$ alone does *not* establish this. Equation (3) explains how vortex stretching can alter enstrophy level sets without destroying the material character of actual vortex tubes.

3 Discovery and independent test

Phase D: At $t = 0$, choose the first closed, watertight $|\boldsymbol{\omega}|^2$ isosurface in the fixed percentile order $65, 75, 85, 92$. Save surface vertices, triangle faces, candidate metadata and SHA-256 seals *before* computing transport. No candidate tuning using tracer outcomes is allowed.

Phase T: Track paired inside/outside initial tracers via RK4 through the evolving velocity snapshots, then independently compare membership relative to the same Eulerian level λ at the final time. Report relative normal speed, crossings and numerical refinement. These observables test an enstrophy feature; full three-dimensional stable/unstable manifold extraction is a distinct unresolved gate.

4 Numerical outcomes (13 time frames; $N = 24$)

Knot	$\Delta E/E$	$\max \lvert\nabla \cdot u\rvert$	Q_{95}/U	Crossings	Stretch res.
3_1	3.64e-10	2.07e-14	0.554	40/218	0.078

All three candidate enstrophy isosurfaces are nonmaterial within their respective *periodic numerical models* at this screening resolution. This is not evidence against the existence of some *other* invariant material surface.

The centered, three-snapshot vortex-stretching identity residuals are exploratory numerical diagnostics. The 6_1 residual is larger and prevents strong qualification even though energy and divergence remain well controlled.

5 Resolution and hard scientific gates

For the 3_1 solver, a two-grid ($N = 24, 32$) velocity comparison gave L^2 discrepancies of 1.17% initially and 1.86% finally. Two grids do not establish asymptotic convergence, and the fit between the core radius and cell width has not been independently calibrated.

- Software control suite: 17 passing tests, including the stationary ABC Beltrami Euler solution and RK4 refinement.
- Time-varying NPZ generator, source hashes and candidate discovery seal: implemented.
- Physical E011 provider-anchor propagation and native independent source agreement: **UNRESOLVED**.
- Spatial/temporal convergence of transport flux and crossing fractions: **UNRESOLVED**.
- Material versus dynamically invariant separatrix manifold: **UNRESOLVED**.
- Framework v1.0.7 official FREEZE/CERTIFY and blind reveal ledger: **NOT EXECUTED**.

6 Falsifiable next gates

1. Replace raw centerlines by every eligible E011 provider anchor; preserve geometry hashes and independent-provider classes. Do not optimize carrier choice on a transport observable.
2. Run $N = 24, 32, 48, 64$ with controlled core width, and CFL/temporal ladders. Check normal flux and paired tracer statistics rather than energy drift alone.
3. Extract actual finite-time hyperbolic/elliptic transport structures or invariant manifolds in a suitable frame; validate material transport under time-varying rigid coordinate transformations.
4. Compare periodic Euler dynamics with an independently implemented, nonperiodic finite-core vortex evolution and with available externally generated snapshot data. No cross-model equivalence is presumed.

7 Reproduction

`run_all.cmd EULER_RUN`; `run_all.cmd EULER_BUILD --force`; `run_all.cmd SELFTEST`. A model-NPZ source sidecar contains the input SHA-256, model parameters, timing, solver diagnostics and science-evidence class. Running the pipeline does not imply SST's physical constants are derived or validated.

References

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[4] G. Haller (2015), "Lagrangian Coherent Structures", *Annual Review of Fluid Mechanics* **47**, 137–162. https://doi.org/10.1146/annurev-fluid-010313-141322.